3DG ENGINE

Build a Game with an Animated Character

A practical start-to-finish tutorial for importing a skeletal character, creating a reusable Character asset, configuring idle-walk-run locomotion, adding full-body actions, testing collision and camera behavior, and exporting the game.

PROJECT	Animated Explorer
CORE LOOP	Explore a small level using idle, walk, run, jump and an interaction action.
OUTPUT	A saved scene plus a reusable .3dgcharacter asset that works in editor Play mode and exported builds.

01 Outcome and production map

This guide builds a compact third-person game around one animated player character. Work in small verified milestones: movement first, animation second, gameplay actions third, and export last.

Milestone	Deliverable	Pass condition
M1 - Project	Folders and saved scene	The scene closes and reopens correctly
M2 - Character	Player Start plus skeletal mesh	Capsule moves and camera follows
M3 - Locomotion	Idle, walk and run	Animation follows actual horizontal speed
M4 - Actions	Jump and interaction or attack	Full-body action completes before movement resumes
M5 - Polish	Material, light, collision and audio	No sliding, clipping or missing assets
M6 - Release	Runtime export	Validation passes and packaged game launches

Recommended project

Create **Animated Explorer**: the player crosses a small obstacle course, runs up stairs, jumps a gap, and activates a goal object. This exercises the complete character pipeline without requiring enemy AI.

Before you begin

- Use a skeletal model with a valid skeleton and embedded or separate animation clips.
- Have at least Idle, Walk and Run. Jump, Fall, Land and an interaction or attack clip are useful additions.
- Keep source assets inside the project Content folder so editor Play and exported builds resolve the same paths.
- Save after every milestone and keep the Console visible during the first test of a new feature.

02 Prepare the project and content

A predictable folder layout prevents broken references and makes the searchable Character Editor pickers useful.

Create these folders in the Assets panel

```
Content/
Scenes/
Assets/Characters/
Assets/Models/
Assets/Animations/
Assets/Materials/
Assets/Audio/
Assets/Particles/
Scripts/
```

Import and verify

1. Import the skeletal model into **Assets/Models**. Supported character containers include FBX, glTF, GLB and DAE.
2. Import textures, then create or open a **.3dgmat** material in Material Maker.
3. Save the level as **Content/Scenes/AnimatedExplorer.scene** before placing gameplay objects.
4. Double-click appropriate assets to confirm their editor panels open. Correct any decode or missing-path errors before character setup.

Orientation check

The engine world uses Y as up. If an imported model lies face-down, correct the model import orientation or the character object's mesh orientation once, then capture that configured object into the reusable character asset. Do not compensate by changing gameplay movement axes.

Asset acceptance checklist

Check	Expected result
Skeleton	Bone hierarchy loads without errors
Clip list	Idle, Walk and Run can be identified
Scale	Character height is close to the capsule height
Material	Albedo, normal and roughness display correctly
Paths	All files live below Content

03 Build a test level

Create a level that makes animation mistakes obvious. Flat ground tests speed; stairs test step handling; a ramp tests slope limits; a wall tests collision and camera retraction.

1. Add a Plane named **Ground**, scale it to about 20 by 20, enable its collider, and select the World Static collision preset.
2. Add four Cube walls with Box colliders. Leave enough room for the camera behind the player.
3. Add a Staircase primitive and a gentle ramp. Keep step height near the player controller's configured step height.
4. Add a Directional Light and tune the sky in World Settings. Enable sun shadows and skylight occlusion for readable indoor areas.
5. Add an Empty object named **Goal** or a visible goal primitive at the far end of the course.

Suggested test geometry

Object	Purpose	Collider
Ground	Continuous walking surface	Plane or Box - World Static
Wall	Movement and camera blocking	Box - World Static
Staircase	Ground probe and step test	Staircase collider
Ramp	Maximum slope test	Box or matching primitive
Goal	Interaction/action destination	Trigger / overlap only

Collision first

A visible mesh does not block the player unless it has a collider. Fix collider shape, collision channel and response before changing animation or movement settings.

Early test

Save the scene, add the player in the next chapter, then walk into every wall and over every stair before adding animation. This isolates controller problems from animation problems.

04 Create the player character

Player Start provides the gameplay capsule and controller. The skeletal model becomes its visual representation; movement remains driven by the controller, not by the mesh orientation.

1. Choose **Add > Gameplay > Player Start**. Keep the object name **PlayerStart**.
2. Place the capsule slightly above the ground. Do not begin with its base intersecting the floor.
3. In Player Controller, use the starting values below and keep third-person mode enabled.
4. Enable Health at 100 / 100 if the game will later include hazards or combat.
5. Assign the Player collision preset. Collectibles and triggers should overlap rather than block it.

Player setting	Starting value	Why
Walk Speed	4.0	Readable locomotion pace
Run Speed	7.0	Clear sprint state
Jump Speed	5.0	Useful for a small obstacle course
Capsule Radius	0.4	Fits a human-sized character
Capsule Height	1.8	Approximate standing height
Camera Distance	5.0	Comfortable third-person framing
Camera Target Height	1.0	Frames the upper body
Maximum Slope	50 degrees	Allows ramps without climbing walls
Step Height	0.35	Supports ordinary stairs

First Play test

Press Play and verify WASD movement, Shift sprint, Space jump, mouse look, camera collision, slope behavior and stairs. Continue only when the capsule behaves correctly with no skeletal model involved.

05 Create a reusable Character asset

The Character Editor gathers the character's mesh, material, capsule, movement, animation, health, AI and script references into one reusable .3dgcharacter file.

1. Select **PlayerStart** and open **Panels > Character Editor**.
2. Choose **Capture Selected**. This copies the current scene object's character-related settings into the editor.
3. Select **Mesh** in the component list. Open the Skeletal Model dropdown and type part of the filename to filter the Content folder.
4. Choose the skeletal model. Open the Material dropdown and select the saved .3dgmat material. Hover an item to inspect its complete path.
5. Review Capsule, Movement, Animation and Gameplay. Use **Apply to Selected** to push the asset back onto PlayerStart.
6. Save as **Content/Assets/Characters/Explorer.3dgcharacter**.

Character Editor layout

Area	Use
Components	Choose Character, Mesh, Capsule, Movement, Animation, Gameplay, AI or Script
Character Preview	Shows the configured capsule and opens live Animation Preview for the selected scene character
Details	Edits the selected component's properties
Capture Selected	Copies a scene instance into the character asset
Apply to Selected	Applies the reusable setup to the selected scene object

Searchable asset lists

The model and material lists scan all subfolders under Content. They show short filenames, search both filename and folder, and expose the full stored path as a tooltip. Use Refresh Asset Lists after importing a new file while the panel is open.

06 Set up idle, walk and run

Locomotion must respond to the character's actual horizontal speed. The animation system selects the appropriate clip while the Player Controller remains responsible for movement and collision.

1. With PlayerStart selected, open **Panels > Animation Preview** or use **Open Animation Preview** from the Character Editor.
2. Confirm the selected object is marked as a skeletal model and verify that every clip plays by itself.
3. Enable **Use Locomotion**. Select Idle, Walk and Run by clip name or index.
4. Set **Walk At** to 0.15 and **Run At** to 3.0 as initial thresholds.
5. Enable looping for locomotion clips and use playback speed 1.0 initially.
6. Apply or capture the finished settings into Explorer.3dgcharacter, then save the scene.

Speed range	State	Expected visual
Below 0.15	Idle	Stationary breathing or stance
0.15 to below 3.0	Walk	Feet advance at the walking pace
3.0 and above	Run	Run or sprint cycle

Animation does not move the controller

The Animation Graph reads movement values; it does not decide where the player moves. Translation follows the controller and world-space collision. This prevents imported model orientation from changing the gizmo or gameplay movement directions.

Reduce foot sliding

- First confirm the clip is authored at the intended pace and the model scale is correct.
- Tune Walk At and Run At so transitions occur near the visual change in stride.
- Then adjust individual state playback speed in small steps. Avoid hiding a major unit-scale error with extreme playback speed.

07 Add airborne and action animation

A production character usually needs Fall and Land states plus one-shot actions such as interact, attack, cast or pickup.

Airborne state graph

From	To	Condition
Idle / Walk / Run	Fall	IsFalling != false
Fall	Land	IsGrounded == true
Land	Idle	Exit time reached and Speed < 0.15
Land	Walk	Exit time reached and Speed >= 0.15

Movement parameters supplied by the engine

Parameter	Type	Meaning
Speed	Float	Horizontal movement speed
VerticalSpeed	Float	Upward or downward velocity
IsGrounded	Bool	Ground probe found a walkable floor
IsFalling	Bool	Gravity is active without floor support
IsFlying	Bool	AI movement intentionally uses Flying mode

One-shot actions

- Use a **full-body, non-layered action** for attacks, rolls or interactions that must hold movement. The character remains stopped until the clip completes, then locomotion resumes.
- Use a **layered action** with a bone-mask root for upper-body actions such as aiming or casting while walking. The legs continue locomotion beneath the action.
- Set sensible fade-in and fade-out values so actions do not snap between poses.
- Add animation events at meaningful times, such as the staff impact, footstep, pickup moment or spell release.

Montage-style rule

An action with no bone mask behaves like a non-layered montage: gameplay locomotion is held until the action finishes. A masked action is layered and does not block movement.

08 Connect gameplay feedback

Animation becomes convincing when movement, collision, audio and effects agree on timing.

Interaction example

1. Place a Goal trigger at the end of the course and configure it as overlap only.
2. Add an interaction action profile to the character. Use a non-layered full-body clip if the character must stop at the goal.
3. Place an animation event named **Activate** at the exact contact moment in the clip.
4. Use a script or trigger action to respond to Activate: play a sound, spawn a particle effect, and mark the level complete.

Logic sketch

```
When player enters Goal trigger:
stop accepting movement for this action
play full-body Interact action
When animation event 'Activate' fires:
play goal audio
spawn goal particle effect
set game state to Victory
When action completes:
return control to locomotion
```

Keep gameplay authoritative

Use the animation event to time feedback, but keep collision, score, health and victory state in gameplay logic. This avoids losing important state changes when an animation is interrupted or replaced.

Optional polish

- Footstep events that choose sounds by surface.
- A small camera shake on landing or interaction impact.
- Dust particles on Land and stronger effects on hard falls.
- A HUD prompt near the goal and a completion message after Activate.

09 Test the complete character

Test in layers. A short repeatable checklist finds most character defects faster than changing several settings at once.

Test	Pass condition	If it fails
Idle	Idle loops without visible drift	Check root translation and loop seam
Walk	State starts above Walk At	Check Speed parameter and threshold
Run	Run begins at the intended pace	Tune Run At and state playback speed
Stop	Character returns cleanly to Idle	Check transition fade and exit rules
Jump / Fall	Airborne state follows the capsule	Check IsGrounded and collider placement
Stairs	Capsule climbs without snapping	Tune step height, ground probe and stair collider
Wall	Player slides or stops, never passes through	Check World Static collision response
Action	Full-body action holds movement until done	Remove bone mask for non-layered behavior
Camera	No clipping through walls or character	Tune probe radius, padding and target height

Debug tools

- Use Animation Preview to inspect clips, parameters, state transitions, action profiles and events.
- Use the physics debug toggle only when checking colliders; turn it off for ordinary Play testing.
- Watch the Console for missing model, material, clip and script paths.
- Use the Character Editor's Capture Selected after a successful adjustment so the reusable asset stays current.

> **Regression habit**
>
> After changing the capsule, movement thresholds or model scale, repeat Idle, Walk, Run, stairs, wall and camera tests. These settings influence one another.

10 Save, validate and export

The final test must use the same saved assets that the standalone runtime will load.

1. Save Explorer.3dgcharacter, then apply it to PlayerStart one final time.
2. Save AnimatedExplorer.scene and close/reopen it. Confirm the scene restores the character, material and animation settings.
3. Run a clean Play session from the beginning. Complete the obstacle course and trigger the goal.
4. Press F7 to export the runtime scene.
5. Press F8 to validate runtime assets. Repair every missing model, material, animation, audio, particle or script path.
6. Build the player target and launch the exported game outside the editor.

Release acceptance

Area	Acceptance result
Startup	Correct saved scene opens
Character	Mesh, material and skeleton load
Movement	Walk, run, jump, slopes and steps work
Animation	Locomotion and full-body action behave correctly
Collision	World blocks player; triggers do not
Camera	Follow and wall retraction are stable
Assets	No absolute-path dependency outside the package

Definition of done

The game is complete when a fresh launch loads the level, the player can traverse it with correct idle-walk-run and airborne animation, the goal action completes with synchronized feedback, validation reports no missing assets, and the standalone player behaves like editor Play mode.

11 Troubleshooting reference

Use the symptom to locate the owning system before adjusting unrelated settings.

Symptom	Likely cause	Repair
Model lies face-down	Import up-axis mismatch	Correct mesh orientation; leave world movement axes unchanged
Movement direction disagrees with gizmo	Visual rotation used as gameplay-axis fix	Restore controller/world axes; rotate only visual mesh orientation
Model moves but stays Idle	Speed parameter or locomotion disabled	Enable locomotion and inspect Speed in Animation Preview
Feet slide	Clip pace, scale or playback mismatch	Verify scale, thresholds, then playback speed
Action moves character	Action has a bone mask	Clear mask for a full-body non-layered action
Character never moves during aiming	Aiming action is full-body	Use an upper-body mask for layered playback
Falls through floor	Missing or incorrect collider	Enable collider and World Static response on floor
Snaps on stairs	Step or probe values do not fit geometry	Tune step height/ground probe and use correct stair collider
Dropdown cannot find asset	File outside Content or list stale	Move/import under Content and click Refresh Asset Lists
Standalone misses model	Unsaved or external path	Save assets below Content and run F8 validation

Final production checklist

- Scene saved under Content/Scenes.
- Character saved as a .3dgcharacter asset and applied to PlayerStart.
- Skeletal model and .3dgmat selected using the searchable lists.
- Idle, Walk and Run verified against Speed.
- Fall and Land verified against grounded movement flags.
- Full-body and layered actions intentionally configured.
- Capsule, stairs, walls and camera collision tested.
- F7 export and F8 validation completed.
- Standalone player launched and tested.

You now have a reusable animated-character pipeline for future projects. Duplicate the Character asset, change the mesh and animation clips, apply it to a new scene object, and retest the same acceptance checklist.